CHAPTER 8

SAMPLING TECHNIQUES

8.1 General Concept of Sampling

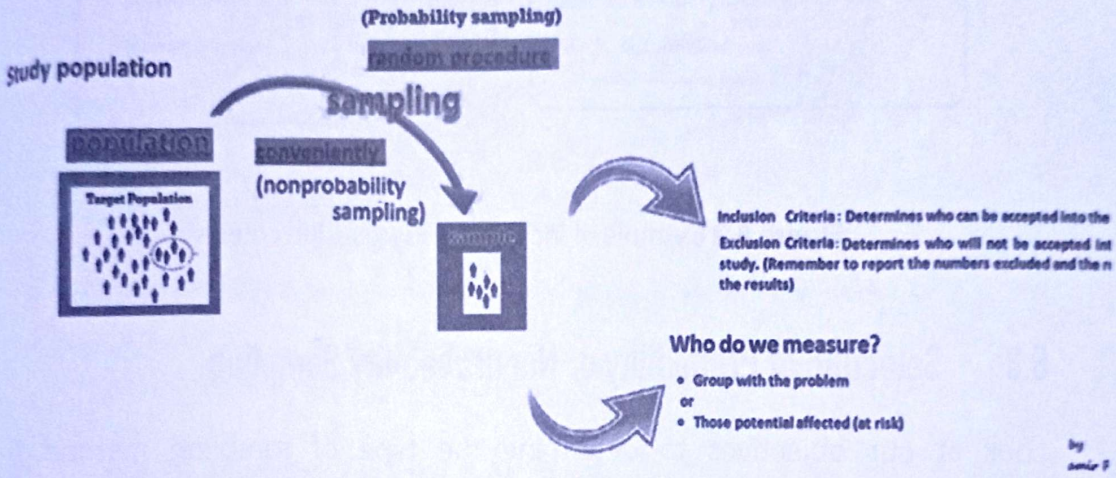


Figure 8.1 General concept of sampling

8.2 Inclusion and Exclusion Criteria Concept

Inclusion and exclusion criteria are used to ensure precision in a study. Researcher must be clear about:

1. Who can be considered as a potential sample.
2. Who cannot be considered as a potential sample.

The criteria are based on such factors as geographical location, religion, age, disease stage, previous treatment, presence of other medical condition and many more. Inclusion criteria is the criteria for including a patient in the study and exclusion criteria is the criteria for excluding patients from the study.

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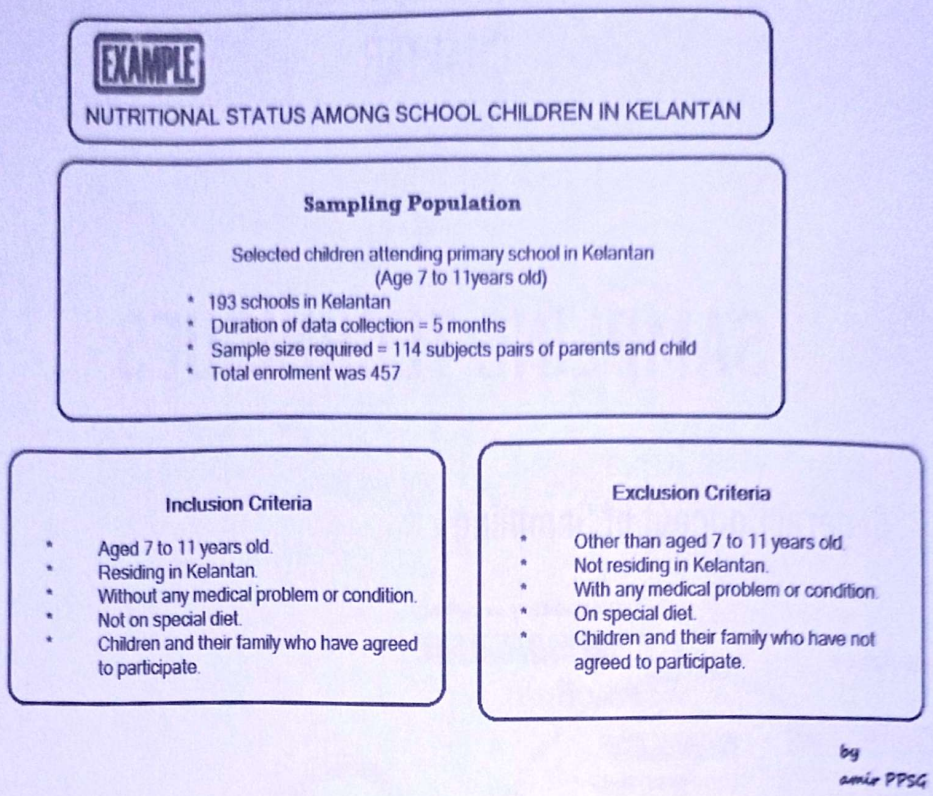


Figure 8.2 Example of inclusion and exclusion criteria

8.3 Selection of Probability or Nonprobability Sampling

Look at our objectives to determine the type of sampling method. If our objectives require you to make inferences (apply the results to the study population), we must select a random sample and use probability sampling methods. If our objectives need a simple and fast answer, we use nonprobability sampling methods. Same situation goes to the case, if we not have a list of study population (sampling frame) then select a convenient sample (nonprobability).

8.4 Sampling Direction

Sampling direction can be summarized as follows:

Reference Population ⇨ Source Population ⇨ Sampling Frame ⇨ Sample

Figure 8.3 Sampling direction

Table 8.1 Study population

Problem	Reference population	Source population	Sampling frame	Sample
High dengue fever prevalence among children under 10 years in Kota Bharu	Children under 10 years with dengue fever in Kota Bharu	Current list of dengue fever among children under 10 years in Kota Bharu	Children with dengue fever under 10 years (2016 to 2017) who fulfill the criteria	Selected sampling frame

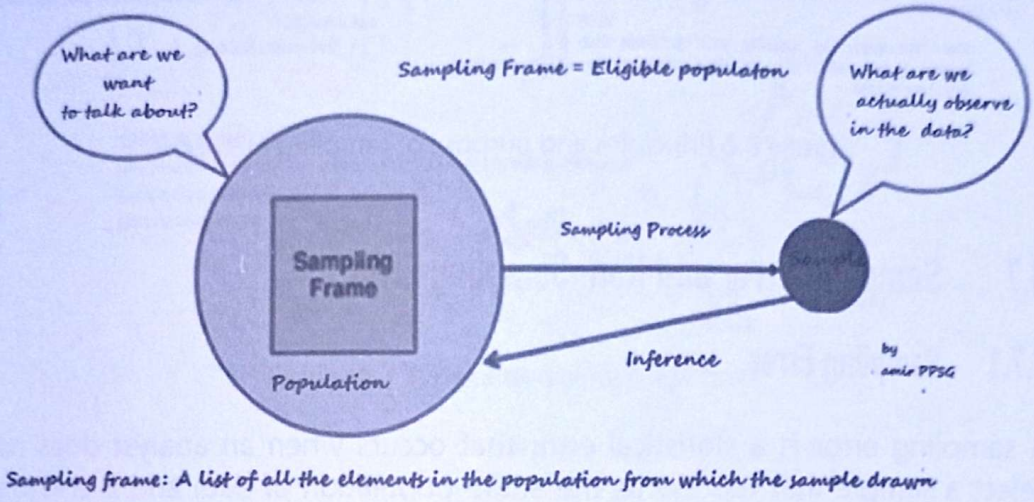


Figure 8.4 Sampling frame process

8.5 Aim of Applying Sampling Method

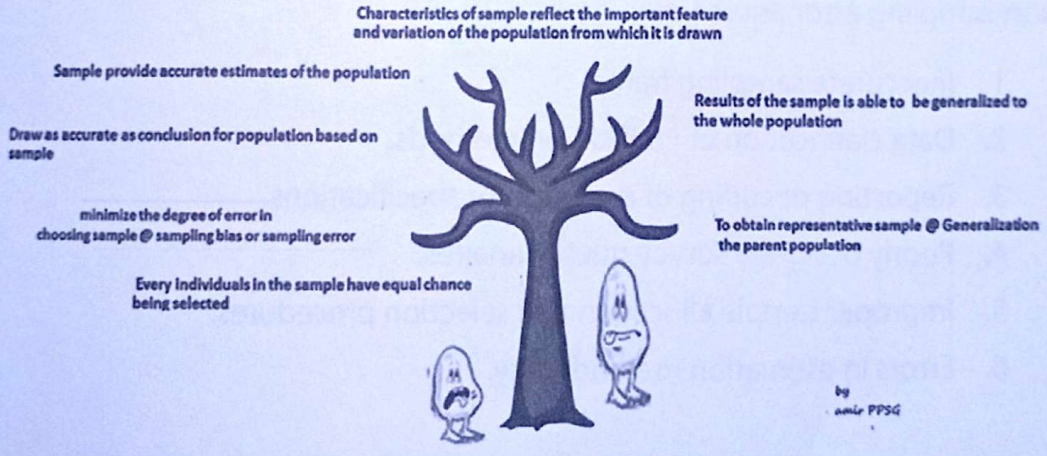


Figure 8.5 Aim of applying sampling method

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8.6 Principles and Purpose of Sampling Method

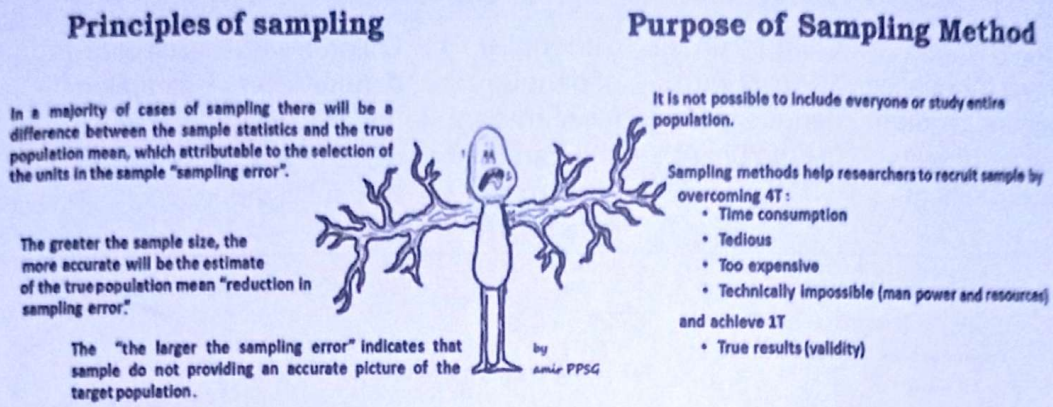


Figure 8.6 Principles and purpose of sampling method

8.7 Sampling Error and Non-Sampling Error

8.7.1 Sampling Error

A sampling error is a statistical error that occurs when an analyst does not select a sample that represents the entire population of data and the results found in the sample do not represent the results that would be obtained from the entire population.

8.7.2 Non-Sampling Error

Non-sampling error arises from:

1. Inaccurate sampling frame.
2. Data clarification or verification methods.
3. Reporting or coding of data, and/or specifications.
4. Poorly designed survey questionnaires.
5. Improper sample allocation and selection procedures.
6. Errors in estimation methodology.

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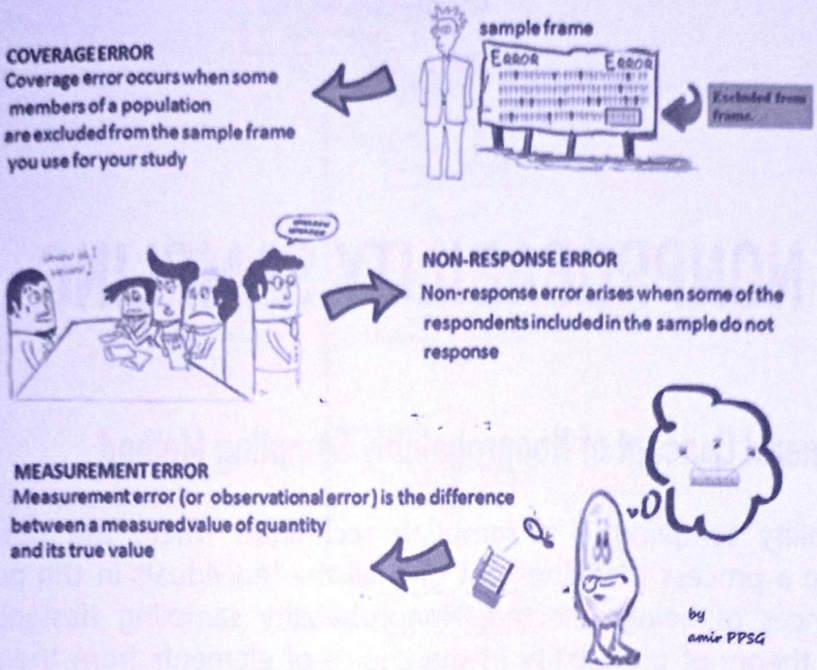


Figure 8.7 Type of non-sampling errors

CHAPTER 9

NONPROBABILITY SAMPLING

9.1 General Concept of Nonprobability Sampling Method

Nonprobability sampling is a sampling technique where the samples are gathered in a process that does not give all the individuals in the population equal chances of being selected. Nonprobability sampling designs do not follow the theory of probability in the choice of elements from the sampling population.

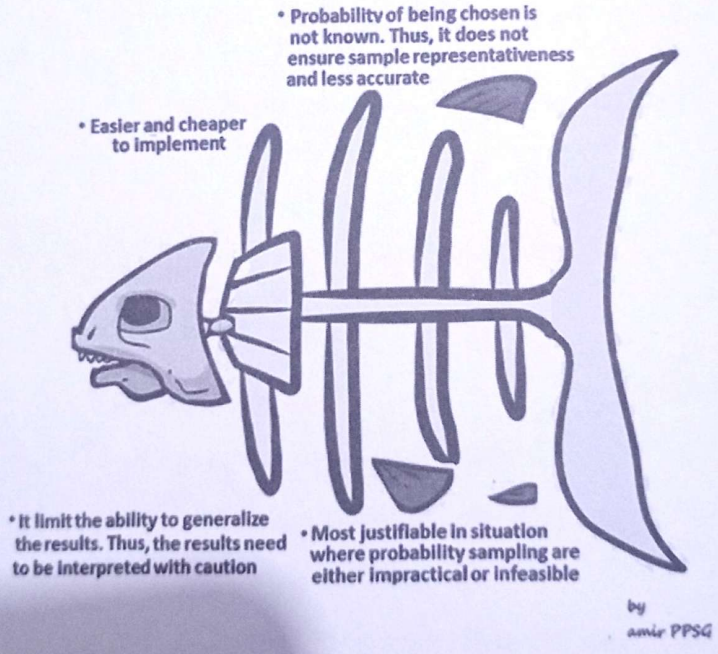


Figure 9.1 Concept of nonprobability sampling method

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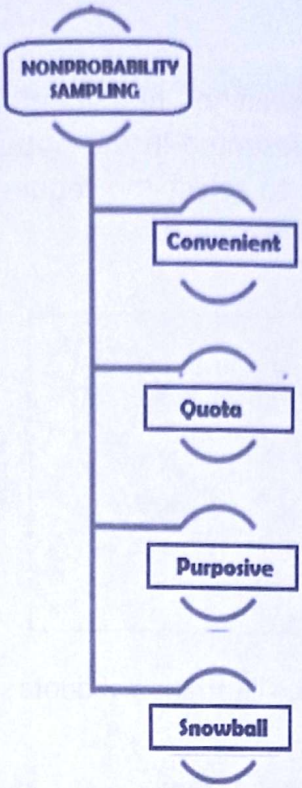


Figure 9.2 Sampling method: Nonparametric sampling

9.2 Convenience Sampling

Convenience sampling is a nonprobability sampling technique where subjects are selected because of their convenient accessibility and proximity to the reseacher. For example,

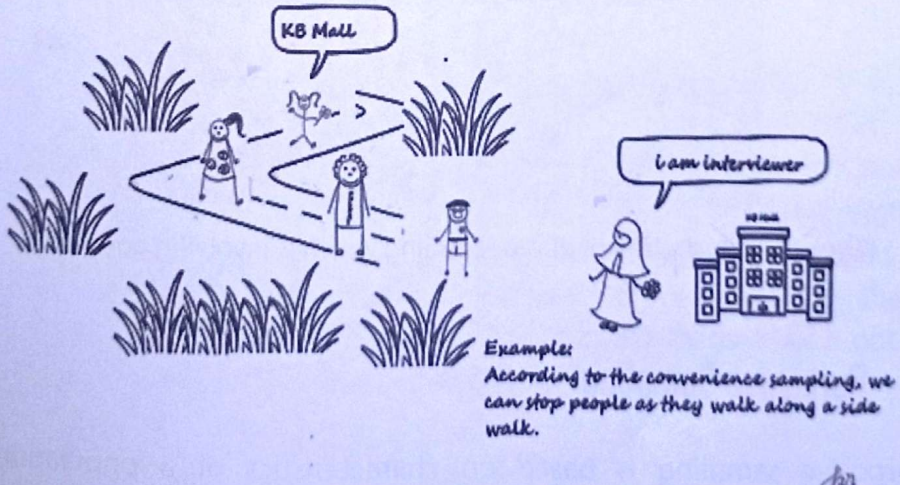


Figure 9.3 Illustration of convenience sampling

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9.3 Quota Sampling

In quota sampling, the researcher first identifies the stratums and their proportions as they are represented in the population. Then convenience or judgment sampling is used to select the required number of subjects from each stratum.

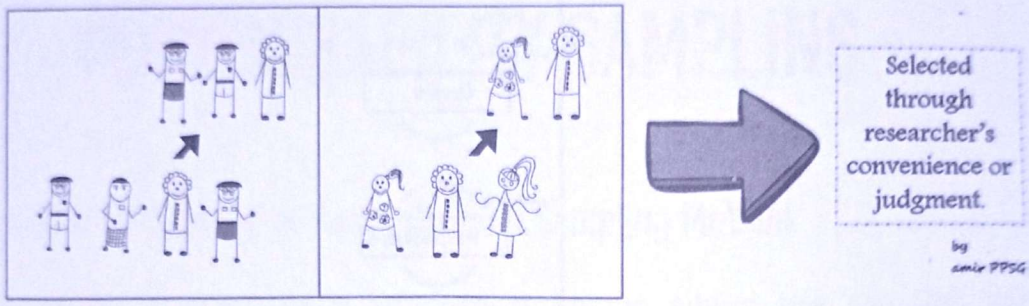


Figure 9.4 Illustration of quota sampling

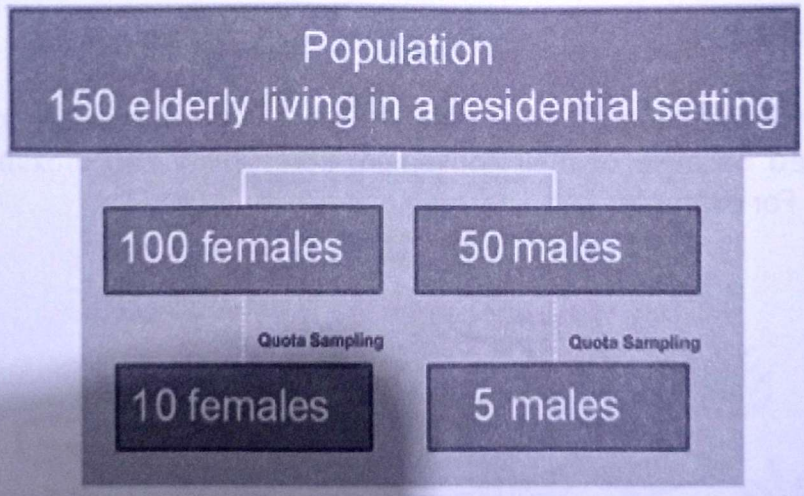


Figure 9.5 Illustration of quota sampling which is involving population

9.4 Purposive Sampling

A purposive sampling is based on characteristics of a population and the objective of the study and focus on the individuals with specified characteristics.

1. Advantage

   Small population of interest, unique event.

2. Disadvantage

   Lack information on most suitable individuals.

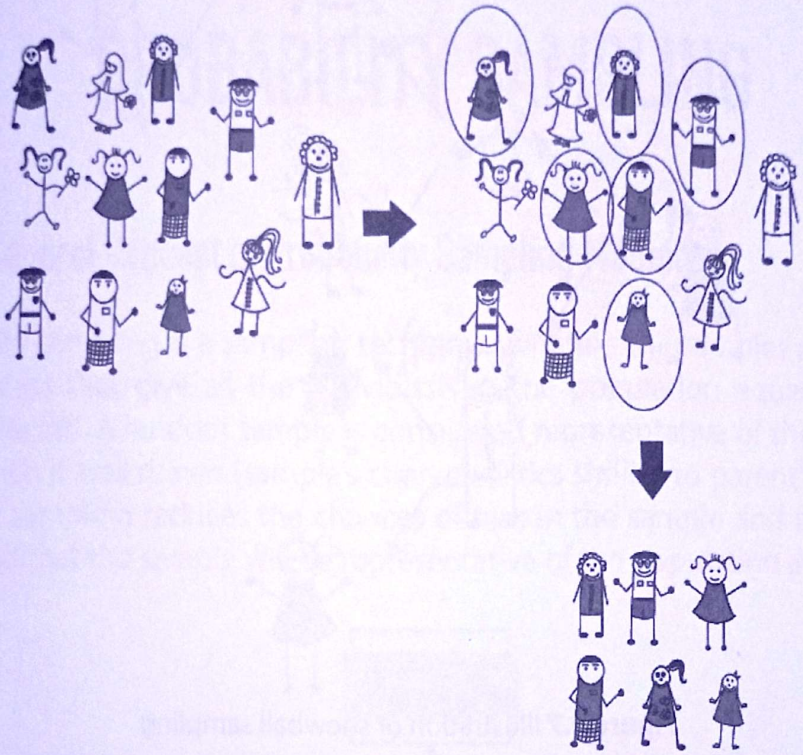


Figure 9.6 Illustration of purposive sampling

9.5 Snowball Sampling

Snowball sampling (also known as chain-referral sampling) is a typically used in qualitative research. Snowball sampling is applicable when subjects with specific characteristics are rare, hard to locate via conventional means and sampling. For the case of snowball sampling, the sampling frame is not applicable. At first, few subjects who meet the criteria are identified. Second stage, these subjects are asked to identify others who have the similar characteristics. The process is continued until adequate number is obtained or researcher reaches saturation of information.

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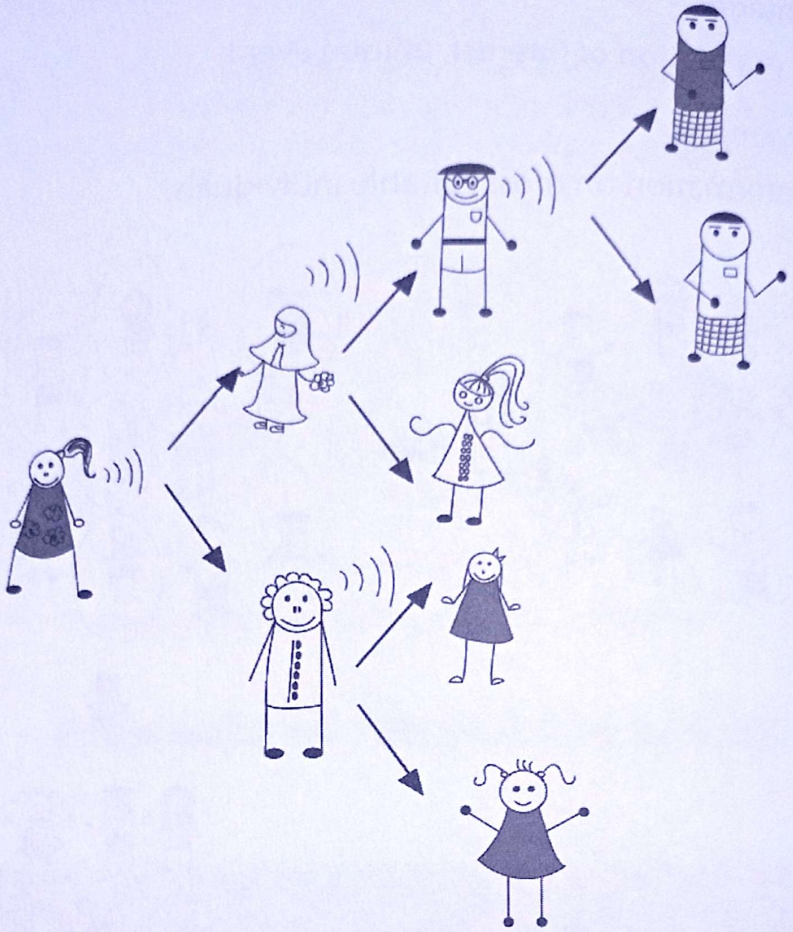


Figure 9.7 Illustration of snowball sampling

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CHAPTER 10

PROBABILITY SAMPLING

10.1 General Concept of Probability Sampling Method

Probability sampling is a sampling technique wherein the samples are gathered in a process that give all the individuals in the population equal chances of being selected. A random sample is considered representative of the population from which it was drawn (sample's characteristics similar to parent population). Random sampling reduces the chances of bias in the sample and increases the likelihood that the sample will be representative of the population at large.

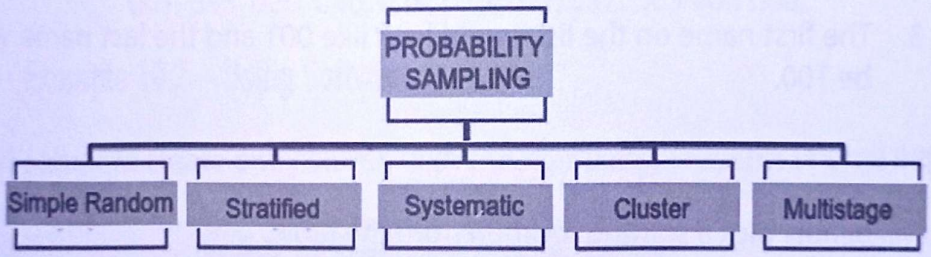


Figure 10.1 Types of probability sampling

10.2 Simple Random Sampling

Suppose we have 100 students registered for a certain awareness program. We want to conduct surveys on 10 students to determine the satisfaction with the program. It is important to choose the 10 students at random. So, we will take a list of all 100 students and use table to randomly pick 10 out of the students from the list, by using random digit table.

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10.2.1 Example 10.1 – Using Random Digit Number

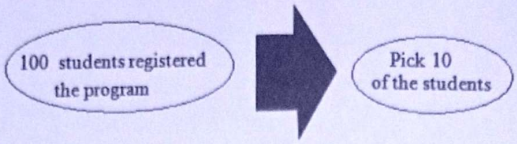


TABLE 1 - RANDOM DIGITS

11164	36318	75061	37674	26320	75100	10431	20418	19228	91792
21215	91791	76831	58678	87054	31687	93205	43685	19732	08468
10438	44482	66558	37649	08882	90870	12462	41810	01806	02977
36792	26236	33266	66583	60881	97395	20461	36742	02852	50564
73944	04773	12032	51414	82384	38370	00249	80709	72605	67497
49563	12872	14063	93104	78483	72717	68714	18048	25005	04151
64208	48237	41701	73117	33242	42314	83049	21933	92813	04763
51486	72875	38605	29341	80749	80151	33835	52602	79147	08868
09756	26360	64516	17071	48478	09610	04638	17141	09227	10606

Figure 10.2 Simple random sampling process

Step I

1. We have the number from 1 to 100 on the list (student number).
2. This means we will need to generate three digit numbers.
3. The first name on the list would look like 001 and the last name would be 100.

Step II

We will blindly pick a starting location from the table.



TABLE 1 - RANDOM DIGITS

11164	36318	75061	37674	26320	75100	10431	20418	19228	91792
21215	91791	76831	58678	87054	31687	93205	43685	19732	08468
10438	44482	66558	37649	08882	90870	12462	41810	01806	02977
36792	26236	33266	66583	60881	97395	20461	36742	02852	50564
73944	04773	12032	51414	82384	38370	00249	80709	72605	67497
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64208	48237	41701	73117	33242	42314	83049	21933	92813	04763
51486	72875	38605	29341	80749	80151	33835	52602	79147	08868
09756	26360	64516	17071	48478	09610	04638	17141	09227	10606



Figure 10.3 Process of sample selection from Random Digits Table (*continued on next page*)

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TABLE 1 - RANDOM DIGITS

11164	36318	75061	37674	26320	75100	10431	20418	19228	91
21215	91791	76831	58678	87054	31687	93205	43685	19732	08
10438	44482	66558	37649	08882	90870	12462	41810	01806	02
36792	26236	33266	66583	60881	97395	20461	36742	02852	50
73944	04773	12032	51414	82384	38370	00249	80709	72605	67
49563	12872	14063	93104	78483	72717	68714	18048	25005	04
64208	48237	41701	73117	33242	42314	83049	21933	92813	04
51486	72875	38605	29341	80749	80151	33835	52602	79147	08

Figure 10.3 (*continued*)

For the further information, follow the following step.

Step III

From our starting point, we will start creating two digit numbers.

750, 613, 767, 426, 320, 751, 001, 043, 120, 418, 192, 289, 179, 221 and so on.

So, the sample are:

001, 043, 029, 046, 028, 056, 047, 097, 063 and 005.

10.2.2 Example 10.2 – Using Software (SPSS)

In SPSS software create one column with the number of students (1 until 100).

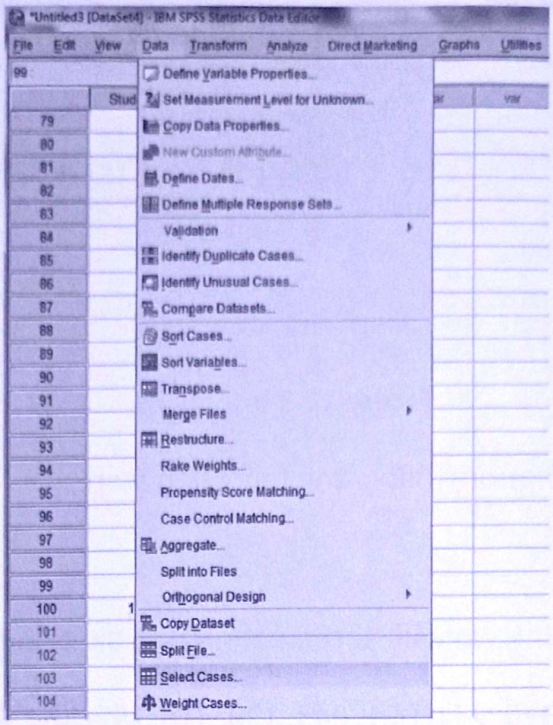
*Untitled3 [DataSet4] - IBM SPSS Statistics Data Editor

File Edit View Data Transform Analyze Direct Marketing Graphs Utilities

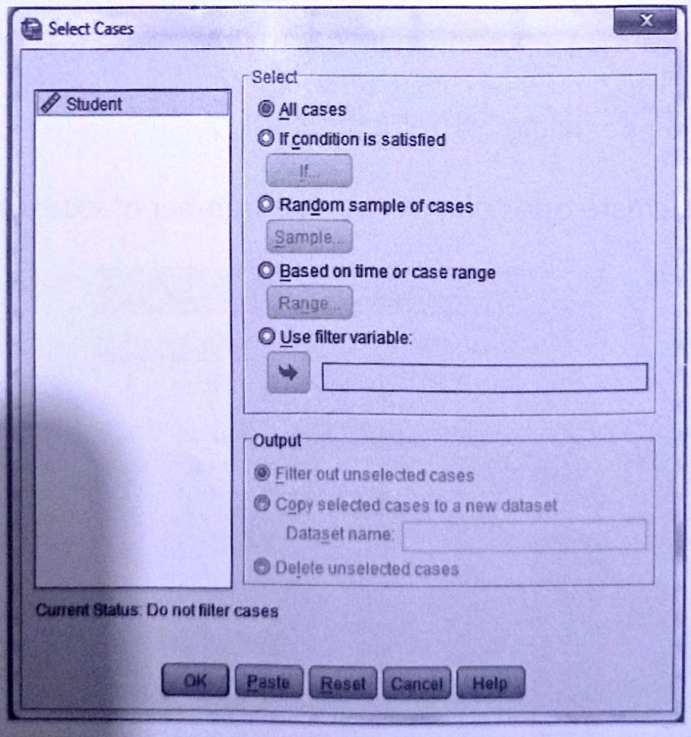
	Student	var	var	var	var	var
1	1.00					
2	2.00					
3	3.00					
4	4.00					
5	5.00					
6	6.00					
7	7.00					
8	8.00					
9	9.00					
10	10.00					
11	11.00					
12	12.00					
13	13.00					
14	14.00					
15	15.00					
16	16.00					
17	17.00					
18	18.00					
...	...					
92	92.00					
93	93.00					
94	94.00					
95	95.00					
96	96.00					
97	97.00					
98	98.00					
99	99.00					
100	100.00					

Select Data, then Select Cases.

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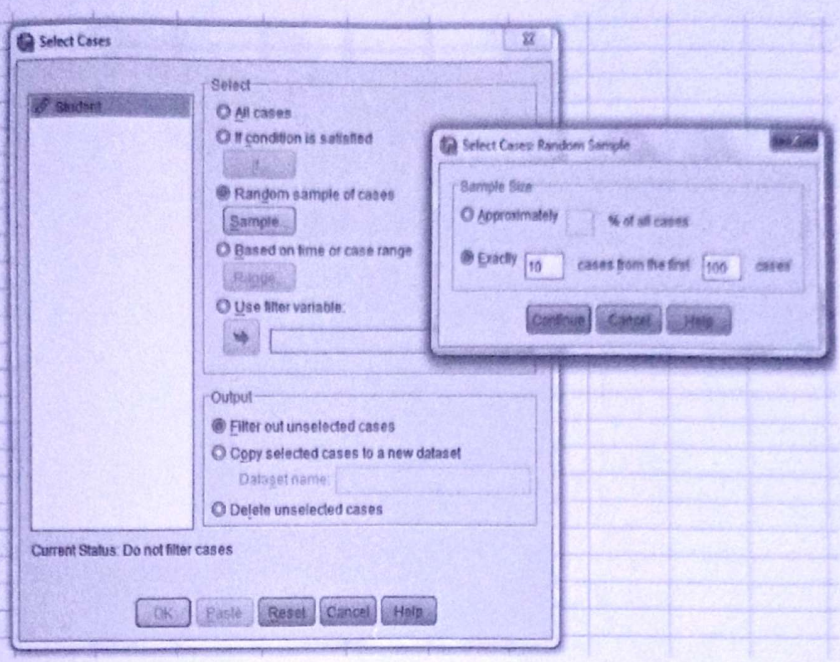


This dialog box will be shown as follows:



Choose Random sample of cases and then click Sample button. Choose exactly 10 cases from the first 100 cases.

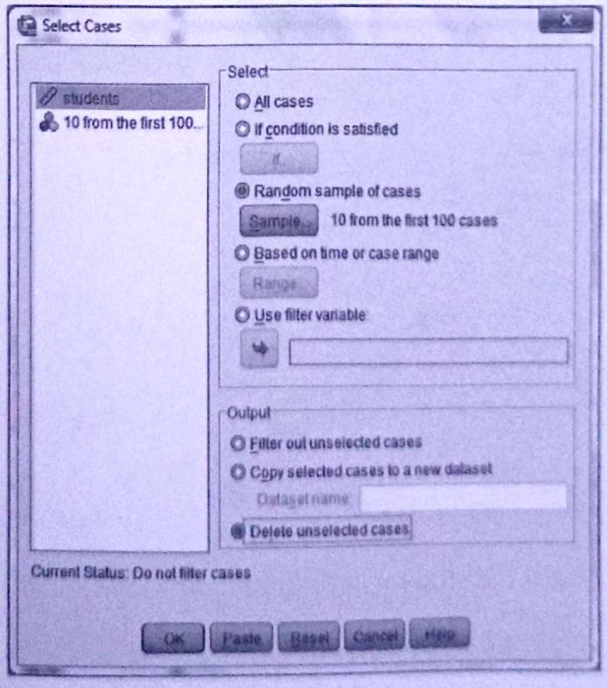
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Click Continue button. At the output section, we can choose:

1. Filter out unselected cases.
2. Copy selected cases to a new dataset.
3. Delete unselected cases.

In our case, we choose to Delete unselected cases.



Then, click OK.

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Output

*Untitled1 [DataSet0] - IBM SPSS Statistics Data Edi

File Edit View Data Transform Analyz

	students	filter_$
1	17	0
2	26	0
3	31	0
4	38	0
5	45	0
6	46	0
7	56	0
8	59	0
9	71	0
10	74	0
11		
12		

10.2.3 Example 10.3

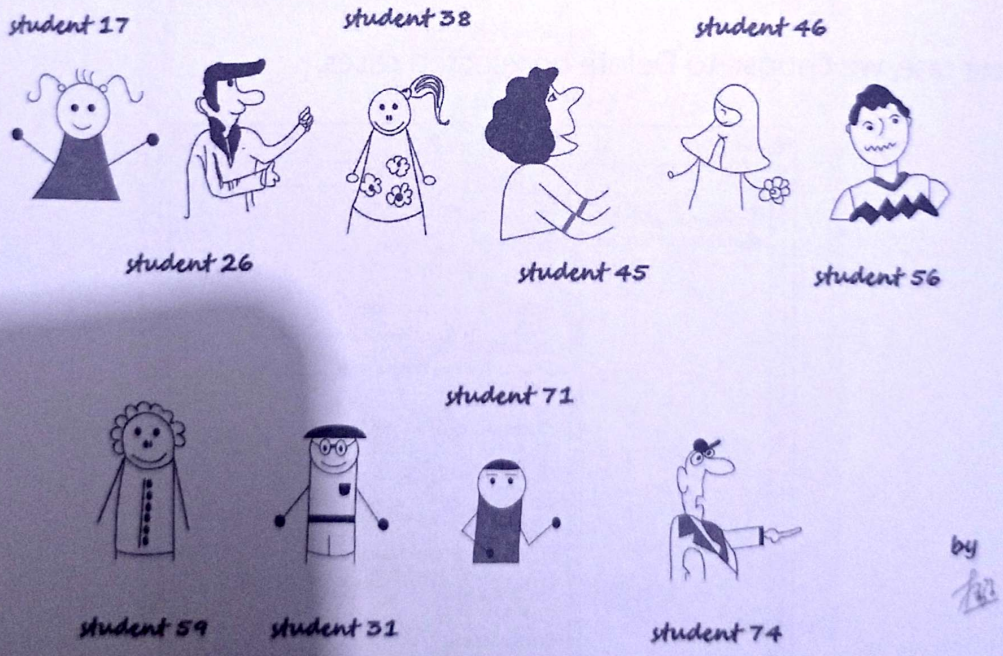


Figure 10.4 Selected student according to SPSS software suggestion

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10.3 Stratified Random Sampling

The population is divided into homogeneous, mutually exclusive groups called strata. Then, independent samples are selected from within each stratum. Different sampling methods or sampling fractions can be used to sample within each stratum.

Note: It is superior to simple random because it reduces sampling error due to stratification.

Examples of common variables (strata):

1. Geographic region : state, district
2. Demographic : age, gender
3. Socioeconomic : income, occupation
4. Clinical : type of treatment, type of health care

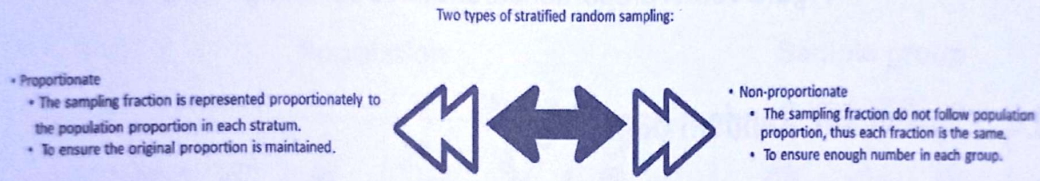


Figure 10.5 Type of stratified random sampling

10.3.1 Example 10.4

10.3.1.1 Case I: Disproportionate Stratified Sample

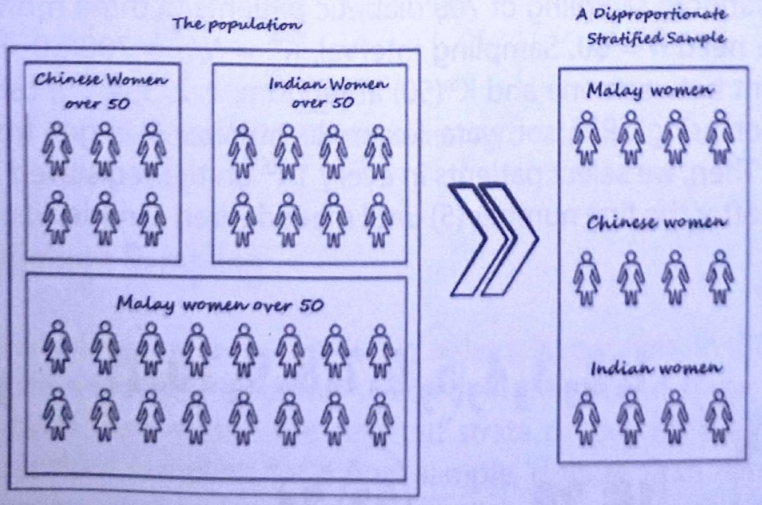


Figure 10.6 A disproportionate stratified sampling

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10.3.1.2 Case II: Proportionate Stratified Sample

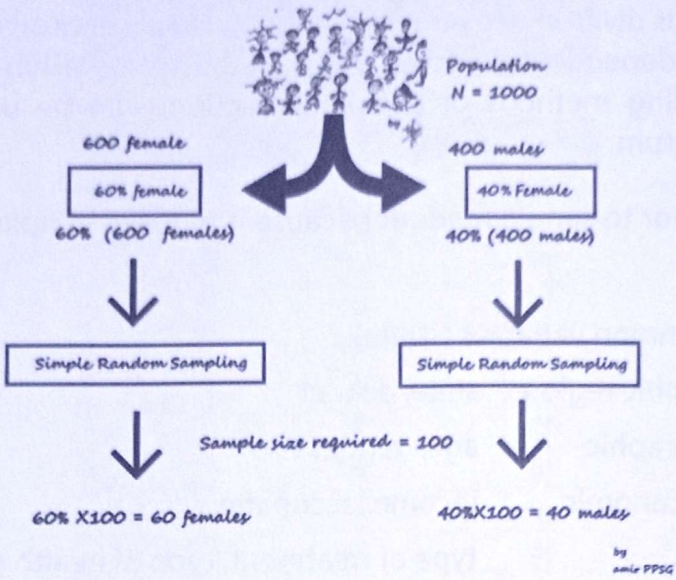


Figure 10.7 A proportionate stratified sampling

10.4 Systematic Random Sampling

In systematic random sampling, the researcher or investigator first randomly picks the first item or subject from the population. Then, the researcher will select each ninth subject from the list.

10.4.1 Example 10.5

Systematic random sampling of 700 diabetic patients in three month $N = 700$. Sample size need $n = 50$. Sampling interval, $K^{th} = N/n = 700/50 = 14^{th}$. Get a starting point between one and $K^{th}(50)$ at random, 1, 2, 3, 4, …, using random digit table or using SPSS software (example number 5 is get from random digit table). Then, we select patients in every 14th on the registered list or clinic attendence after the first number (5) until meet desired sample size ($n = 50$).

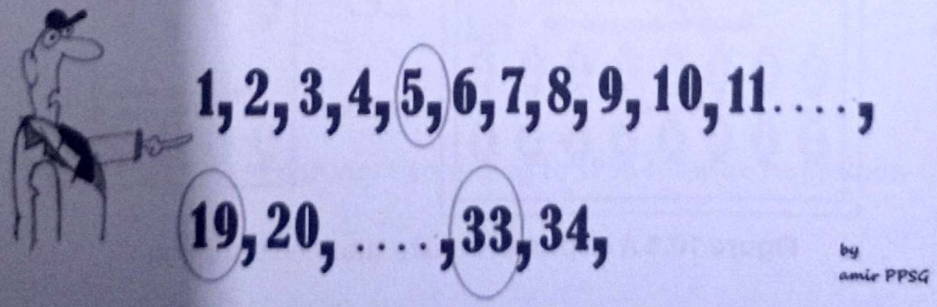


Figure 10.8 Systematic random sampling procedure

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10.5 Cluster Sampling

Cluster sampling is a technique in which clusters of participants that represent the population are identified and included in the sample. Cluster involves cluster of participants that represent the population are identified and included in the sample.

10.5.1 Example 10.6

Steps:

1. A cluster sample is obtained by selecting all individuals within a randomly selected collection or group of individuals.
2. Since we want a random sample of size six, we just select two of the clusters. We would number the clusters 1–6 and use technology to randomly select two random cluster numbers. It might look something like this:

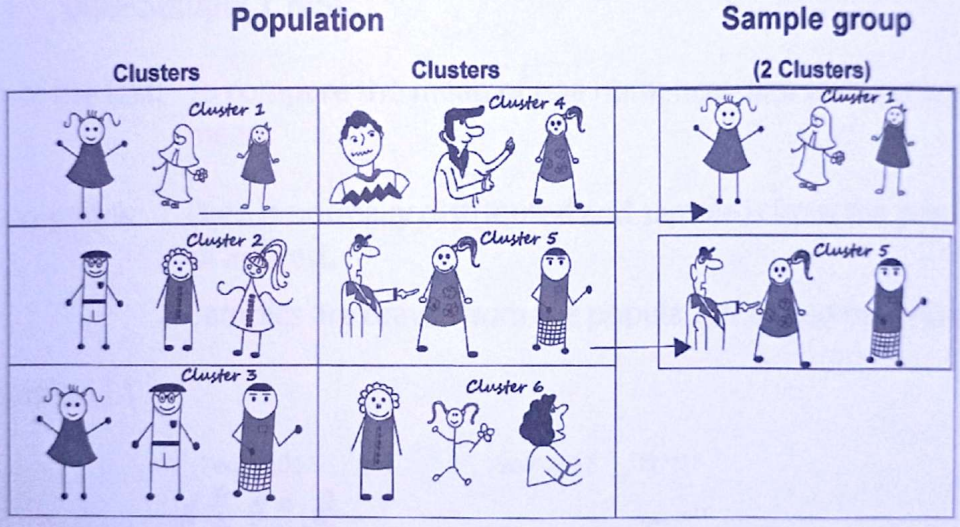


Figure 10.9 Cluster sampling procedure

10.6 Multistage Sampling

Multistage sampling requires sampling in at least two stages. In the first stage, strata or clusters are identified and selected. In the second stage, population units are picked from within the selected strata or clusters using simple or systematic random sampling for a final sample. If more than two stages are used, the process of choosing population units within clusters continues until meet final sample.

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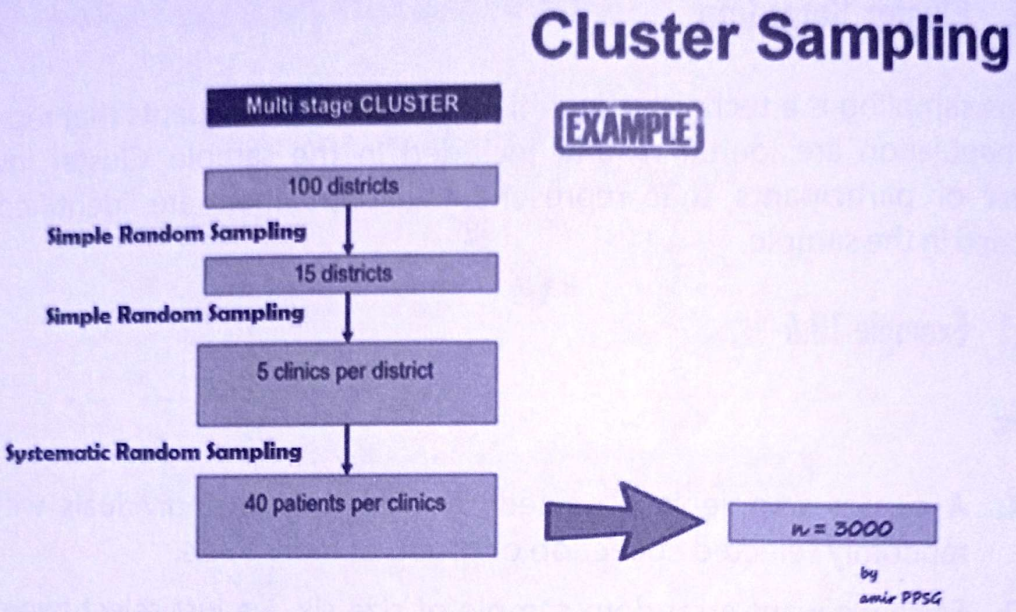


Figure 10.10 Example of multistage cluster sampling

SAMPLE SIZE CALCULATIONS

(STUDY DESIGN BASED)

USING

PS SOFTWARE

AND

SAMPLING SELECTION

WAN MUHAMAD AMIR W AHMAD
NORSAMSU ARNI SAMSUDIN
MUNIRAH MOHD ADNAN
ADAM HUSEIN

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